

RANDOLPH AFB, TX, USA

WMO: 722536

Lat: 29.533N

Lon: 98.262W

Elev: 222

StdP: 98.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12911

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	0.0	-11.1	1.5	5.0	-8.7	1.8	7.4	11.7	9.3	10.8	11.6	4.0	330	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	37.9	23.2	37.0	23.2	35.9	23.3	25.8	30.6	25.6	30.4	25.2	30.1	4.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.8	26.6	24.6	20.1	26.5	24.1	19.5	26.4	81.3	30.7	79.5	30.6	77.9	30.0	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.8	8.7	7.9	-5.5	39.8	2.2	1.9	-7.1	41.2	-8.4	42.3	-9.6	43.4	-11.2	44.8
			-6.8	26.9	1.8	0.7	-8.1	27.3	-9.1	27.7	-10.1	28.1	-11.4	28.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	11.3	14.0	17.0	20.5	24.8	28.0	29.3	29.6	26.5	21.6	16.1	12.0
	DBStd	7.63	5.12	5.63	4.90	3.87	3.05	2.08	1.69	1.77	2.86	4.33	4.99	5.21
	HDD10.0	127	46	24	8	1	0	0	0	0	0	1	10	37
	HDD18.3	796	223	140	84	22	2	0	0	0	1	21	98	206
	CDD10.0	4116	86	136	223	317	458	541	598	607	496	360	193	100
	CDD18.3	1745	5	18	42	88	202	291	340	349	247	122	31	10
	CDH23.3	18201	32	128	259	704	1781	3189	4061	4249	2484	1064	199	52
CDH26.7	8774	2	29	52	217	731	1586	2213	2357	1192	364	28	4	
Wind	WSAvg	3.9	3.9	4.0	4.3	4.4	4.3	4.1	3.8	3.4	3.1	3.6	3.6	3.7
Precipitation	PrecAvg	861	52	48	64	64	105	92	71	59	85	102	64	55
	PrecMax	1289	145	142	162	180	217	317	373	207	271	515	270	340
	PrecMin	389	0	2	0	5	3	2	4	3	2	0	1	4
	PrecStd	272	42	40	37	47	65	75	87	58	56	109	67	67
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	30.2	30.2	33.8	35.9	38.8	39.0	39.1	37.8	33.6	29.2	27.2
		MCWB	16.4	17.2	18.6	20.9	23.1	23.7	23.0	23.2	22.1	22.0	20.8	18.5
	2%	DB	23.9	26.8	28.0	31.1	34.0	36.7	37.7	37.9	35.8	32.0	27.4	24.9
		MCWB	16.0	17.3	18.8	20.5	23.3	23.5	23.1	23.4	22.8	22.2	19.6	18.2
	5%	DB	22.0	24.7	26.3	29.0	32.4	35.2	36.7	37.0	34.1	30.4	25.9	22.6
		MCWB	16.0	17.1	19.0	20.3	23.2	23.6	23.2	23.3	22.6	21.5	19.1	17.6
	10%	DB	20.0	22.4	24.3	27.4	30.8	33.7	35.3	35.9	32.6	28.8	23.9	20.9
		MCWB	15.2	16.7	18.0	19.7	23.0	23.9	23.5	23.4	22.8	21.1	18.7	17.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.6	21.2	22.4	24.4	26.1	26.5	26.2	26.2	25.9	25.5	23.4	22.0
		MCDB	22.1	24.0	26.2	28.5	30.7	31.8	31.0	30.6	29.4	28.6	25.9	23.7
	2%	WB	19.4	20.2	21.4	23.3	25.2	25.8	25.7	25.6	25.3	24.6	22.3	20.6
		MCDB	21.5	22.8	24.5	27.4	30.1	30.9	30.6	30.5	29.5	27.8	24.9	22.6
	5%	WB	18.0	19.5	20.6	22.5	24.6	25.5	25.2	25.2	24.8	23.9	21.1	19.7
		MCDB	20.2	22.3	23.4	26.3	29.2	30.4	30.4	30.4	29.1	27.2	23.7	21.8
	10%	WB	16.5	18.6	19.9	21.8	24.0	25.0	24.9	24.9	24.4	23.1	20.2	17.9
		MCDB	18.9	21.2	22.7	25.2	28.3	29.9	30.0	30.2	28.7	26.5	22.9	20.0
Mean Daily Temperature Range	5% DB	MDBR	12.4	12.0	11.7	12.2	10.9	11.1	12.0	12.2	11.6	12.2	11.7	11.7
		MCDBR	14.8	14.1	12.8	13.9	12.3	13.1	13.8	13.7	13.4	12.7	12.8	13.6
		MCWBR	8.7	7.8	6.4	6.3	4.1	3.0	2.9	2.9	3.6	5.0	7.0	8.3
	5% WB	MCDBR	11.4	10.6	10.2	10.2	10.2	10.3	11.5	11.6	10.2	9.6	10.4	11.0
		MCWBR	8.2	7.3	6.5	5.5	4.3	3.3	3.0	3.0	3.4	5.0	6.9	8.4
Clear-Sky Solar Irradiance	taub	0.330	0.334	0.360	0.401	0.447	0.429	0.438	0.423	0.415	0.373	0.362	0.318	
	taud	2.487	2.482	2.421	2.323	2.238	2.343	2.353	2.375	2.352	2.456	2.417	2.533	
	Ebn at Noon	907	934	927	895	850	860	851	863	860	880	860	904	
	Edh at Noon	91	100	113	129	141	127	125	121	120	102	97	84	
All-Sky Solar Radiation	RadAvg	3.01	3.52	4.47	5.32	5.76	6.58	6.63	6.33	5.20	4.32	3.24	2.67	
	RadStd	0.34	0.64	0.44	0.52	0.56	0.42	0.57	0.36	0.52	0.57	0.36	0.34	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86

Nomenclature: See separate page